

Differentiable functions – 5

Convexity as a method: Jensen, Hermite–Hadamard,
Karamata

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Theoretical minimum.

Theorem 1 (Jensen). For a convex f and weights $\lambda_i \geq 0$ with $\sum \lambda_i = 1$, $f(\sum \lambda_i x_i) \leq \sum \lambda_i f(x_i)$; equality for a strictly convex f forces all x_i with $\lambda_i > 0$ to coincide. The integral form is $f(\int g) \leq \int f \circ g$ for a probability measure.

Theorem 2 (Hermite–Hadamard). For a convex f on $[a, b]$,

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx \leq \frac{f(a)+f(b)}{2},$$

and each of the two inequalities characterises convexity.

Theorem 3 (Karamata). If (x_i) majorises (y_i) , that is the decreasing rearrangements satisfy $\sum_{i \leq k} x_i \geq \sum_{i \leq k} y_i$ for every k with equality at $k = n$, then $\sum f(x_i) \geq \sum f(y_i)$ for every convex f .

Fact 1. The tangent line trick: to prove $\sum f(x_i) \geq c$ under a constraint $\sum x_i = s$, compare f with its tangent line at $\frac{s}{n}$; the inequality follows termwise whenever the graph stays above that line.

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1. Prove Jensen's inequality in the discrete and in the integral form, and deduce the inequalities between the power means and Hölder's inequality.
 2. Prove the Hermite–Hadamard inequality and show that either half of it, assumed for all subintervals, already implies convexity.

3. Prove Karamata's inequality and use it to prove that in a triangle $\cos A + \cos B + \cos C \leq \frac{3}{2}$ and $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$.
4. (tangent line trick) Let $x, y, z > 0$ with $x + y + z = 3$. Prove that $\frac{x}{x^2+1} + \frac{y}{y^2+1} + \frac{z}{z^2+1} \leq \frac{3}{2}$, and explain exactly which property of the tangent line at 1 makes the argument work.
5. (RMO) Let $f: [0, \infty) \rightarrow [0, \infty)$ be convex and differentiable, and suppose there is $s > 1$ with $x^s f'(x) \geq (f(f(x)))^s$ for all $x \geq 0$. Prove that $f(x) \leq x$ for all $x \geq 0$.